Zigang Li

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Education

Rotman School of Management, University of Toronto
Ph.D. in Finance, 2021 – 2027 (expected)

Columbia Business School
M.S. in Financial Economics, 2018 – 2020

Tsinghua University
B.E. in Industrial Engineering, Minor in Finance, 2014 – 2018
Outstanding Graduate

Research Interests

Big Data and AI in Finance; Information in Financial Markets

Job Market Paper

Data Technology and Market Efficiency

Publications

Understanding Rationality and Disagreement in House Price Expectations
(with Stijn Van Nieuwerburgh and Wang Renxuan)
***Review of Financial Studies**, February 2026, vol. 39 (2), pp. 297–342*
(Editor's Choice)
Conference presentations: AREUEA National Meeting, the Sixth CEIBS Finance and Accounting Symposium, the NBER Behavioral Finance Meeting, the Hoyt Conference, the Finance Down Under Conference, the Adam Smith Workshop, WFA, EFA

Working Papers

The Limits of AI Trading
(with Winston Wei Dou, Itay Goldstein, and Liyan Yang)
Conference presentations: NBER SI Financial Market Structure, WFA, EFA, CICF

Inelastic Demand at the Extensive and Intensive Margins
(with Xindi He and Allen Hu)
Conference presentations: NBER SI Financial Market Structure, HEC-McGill Winter Finance Workshop

Work In Progress

An Agentic Approach to Estimating Investor Demand and Beliefs
(with Allen Hu)

Research Experience

Research Assistant for Professor Liyan Yang	2022 – 2023
Research Assistant for Professor Stijn Van Nieuwerburgh	2019 – 2021

Teaching Experience

Rotman School of Management, University of Toronto
Teaching Assistant

Applications of Machine Learning in Finance Fall 2024, Winter 2025
(Undergraduate, by Professors Goutham Gopalakrishna and Raymond Kan)

Capital Market Theory Winter 2024, Winter 2025, Winter 2026
(Undergraduate, by Professors Goutham Gopalakrishna and Kevin Wang)

Management Analytics Practicum Winter 2024
(Master of Management Analytics, Operations Management and Statistics Dept., by Professor Dmitry Krass)

Columbia Business School
Teaching Assistant

Fintech: Consumer Financial Services Spring 2020
(EMBA, by Professor Stephen Zeldes)

Professional Services

Referee: Journal of Economic Dynamics and Control, Journal of Financial Markets

Other Participation

Summer School on Structural Estimation in Corporate Finance 2023

Honors, Grants, and Fellowships

Canadian Securities Institute Research Foundation PhD Award ($35,000)	2026–2027
University Doctoral Excellence Award ($10,000)	2026–2027
TD Management Data and Analytics Lab Research Grant ($5,000)	2026
Ontario Graduate Scholarship (OGS) ($15,000)	2025–2026
TD Management Data and Analytics Lab Research Grant ($6,000)	2025
Tsinghua University Outstanding Undergraduate Thesis Award	2018
Tsinghua University Academic Excellence Scholarship	2014–2017
First Prize, 32nd China National College Physics Contest	2015

References:

Liyan Yang (Chair)
Professor of Finance
Rotman School of Management,
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Stijn Van Nieuwerburgh
Earle W. Kazis and Benjamin Schore Professor of Real Estate and Professor of Finance
Graduate School of Business,
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Itay Goldstein
Joel S. Ehrenkranz Family Professor, Professor of Finance, and Professor of Economics
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Abstracts

Data Technology and Market Efficiency
(Job Market Paper)

How will advances in data technology affect market equilibrium and informational efficiency when adoption by investors is uneven? I show that data technology development and dissemination improve asset managers' stock-picking ability, especially for those with larger shares of data-science labor. Using U.S. institutional holdings and employment records, I estimate information production across investors and a demand system with heterogeneity in private information and learning from prices. I estimate investors' private information from holdings' forecasting power for future payoffs beyond public information and prices. Data technology developments are captured by increases in data scientist productivity. Estimates show that better-informed investors have more elastic demand, while learning from prices makes demand 61 percent less elastic at the median informative stock. A counterfactual advance calibrated to more than a decade of data scientist productivity growth raises price informativeness by 26 percent, more for small stocks. New technology adopters crowd out other investors, increasing market segmentation as stocks' investor bases diverge. Aggregate demand becomes more inelastic for 47 percent of stocks.

Understanding Rationality and Disagreement in House Price Expectations
(Published in Review of Financial Studies)

Professional house price forecast data are consistent with a rational model where agents must learn about the parameters of the house price growth process and the underlying state of the housing market. Slow learning about the long-run mean generates overreaction to forecast revisions and a modest response of forecasts to lagged realizations. Heterogeneity in signals and priors about the long-run mean helps the model account for cross-sectional dispersion in forecasts. Introducing behavioral frictions helps improve the model's predictions for short-horizon overreaction and dispersion. Using a cross-section of forecasters and a term structure of forecasts are crucial for inference.

The Limits of AI Trading

This paper develops a framework for AI-human interactions in financial markets, in which AI investors learn the price formation process through Q-learning while human investors form beliefs through level-k thinking, capturing how the two types differ in learning about the market environment. We show that for any human behavior, AI trading converges to a rational strategy that correctly understands human trading and the true price function. This convergence facilitates identifying two limits humans place on AI performance: AI's information advantage is bounded by human information, and human sophistication limits AI sophistication.

Inelastic Demand at the Extensive and Intensive Margins

We decompose investor demand into its extensive margin—initiating or liquidating a position—and intensive margin—scaling an existing position. Extensive demand is economically large, accounting for 40% of institutional and 80% of retail flows, and its estimated price elasticity is closer to zero and often positive. Extensive demand is primarily driven by past returns and attention, whereas intensive demand reflects fundamentals. To account for these findings, we develop a Grossman–Stiglitz-style model of inelastic markets in which extensive flows act as unobservable endogenous demand shocks and investors learn from prices based on their subjective expectations

about these flows. Learning from prices leads to a more inelastic, and even upward-sloping, extensive demand curve. Embedding the model in an asset demand system, we structurally estimate both elasticities under alternative specifications of flow expectations and find that extrapolative expectations best fit the data. Reduced-form evidence and counterfactual analysis both show that as the share of extensive demand rises, prices, volatility, and firm-specific price informativeness increase, while market-wide informativeness declines.